



MRC
Biostatistics
Unit



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Adaptive Methods in Clinical Research

Lecture 8: Estimation for Adaptive Designs

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Outline

Introduction

Point estimation for ADs

Case study: MUSEC trial

Confidence intervals

Case study (continued)

Simulation study

Discussion

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Discussion

- Appropriate estimation of treatment effects is a key part of trial validity
- The question of estimation of treatment effects in an adaptive clinical trial has received comparatively less attention
- One key issue with estimation after an adaptive clinical trial is that the conventional **point estimators** can be prone to **bias**.
 - ▶ FDA (2019): *“A systematic tendency for the estimate of treatment effect to deviate from its true value” . . . “Biased estimation in adaptive designs is currently a less well-studied phenomenon than Type I error probability inflation”*
 - ▶ ICH E20 (2025): *“Use of conventional analysis methods that would apply in non-adaptive designs usually lead to . . . [a] biased treatment effect estimate.”*

Introduction

- Capturing uncertainty through the use of **confidence intervals (CIs)** is also essential to aid interpretation
- Standard CIs for adaptive designs may not have desirable properties
 - ▶ Can have incorrect **coverage** (probability that the CI contains the true unknown treatment effect of interest)
- ICH E20 (2025): *“The primary analysis should also support calculation of accurate measures of uncertainty such as confidence intervals with targeted coverage probabilities.”*
- FDA (2019): *“confidence intervals for the primary and secondary endpoints may not have correct coverage probabilities for the true treatment effects . . . [and thus] . . . confidence intervals should be presented with appropriate cautions regarding their interpretation”*
- EMA (2007): *“methods to . . . provide confidence intervals with pre-specified coverage probability are required”*

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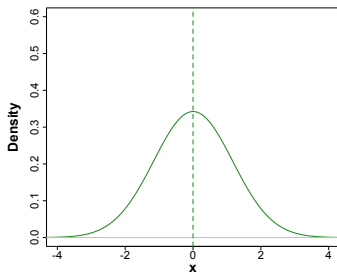
Case study (continued)

Simulation study

Discussion

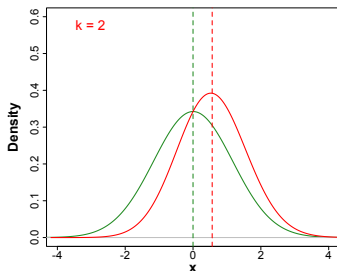
Toy example

- Suppose outcome of experimental treatment follows a standard normal distribution
- What happens when we select the best-performing (i.e. treatment with highest observed mean) of k such treatments?



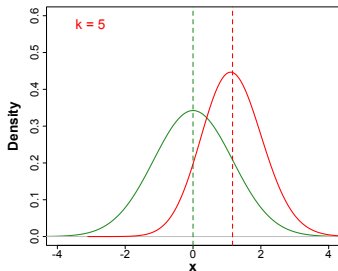
Toy example

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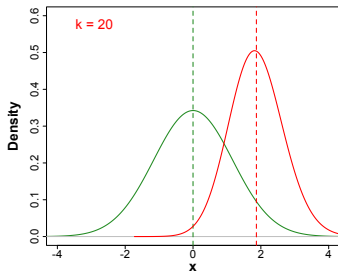
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Why is biased estimation a potential problem?

- Reporting substantially biased estimates for a primary outcome measure following an adaptive design can result in poor decisions
- Concern about over- or under-estimation of treatment effects affecting further research
 - ▶ In a phase II trial, ineffective treatments with exaggerated effects may be wrongly selected for further investigations in phase III trials or potentially effective treatments may not be pursued further when their effects are underestimated
- Impact on health economic analyses
 - ▶ Patients may be penalised when a treatment is not funded based on an underestimate of cost-effectiveness, or resources may be wasted based on an overestimate of cost-effectiveness.

- Mean bias of an estimator $\hat{\theta}$ of the parameter θ is

$$\text{bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

- An estimator that has mean bias identically equal to 0 (for all possible values of $\theta \in \Theta$) is said to be *mean-unbiased*
- The *mean squared error* (MSE) is

$$\text{mse}(\hat{\theta}) = E(\hat{\theta} - \theta)^2$$

- Can be decomposed into the sum of the variance and the bias squared:

$$E(\hat{\theta} - \theta)^2 = \text{var}(\hat{\theta}) + [\text{bias}(\hat{\theta})]^2$$

Desirable criteria for point estimators

- Adequately reflects the adaptive design used
- No or small bias
- Low MSE (reflecting a favourable bias-variance trade-off)
- Is easily computable
- ICH E20 (2025): *“Sponsors should evaluate bias and variability of treatment effect estimates, including measures such as the mean squared error. In the trade-off between bias and variance, the expectation is generally for limited to no bias in the primary estimate of the treatment effect.”*

Unbiased and bias-reduced estimators

- Lots of methods to remove or reduce the bias in the MLE
- Two broad classes:
 1. Unbiased estimators
 2. Bias-reduced estimators
- Also distinguish between *conditionally* and *unconditionally* unbiased estimators:
 - ▶ *Unconditionally* unbiased if it is unbiased when averaged across all possible realizations of an adaptive trial
 - ▶ *Conditionally* unbiased if it is unbiased only conditional on the occurrence of a subset of trial realizations
 - ▶ E.g. one might be interested in an estimator only conditional on a particular arm being selected at an interim analysis

- Sensible way of choosing which unbiased estimator to use is to find the unbiased estimator with the smallest possible variance
- Such an estimator is called the *uniformly minimum variance unbiased estimator* (UMVUE)
- Use the Lehmann-Scheffé theorem:
 - ▶ Suppose S is a minimal sufficient and complete statistic, and U is an unbiased estimator
 - ▶ Then the estimator $\hat{U} = E(U|S)$ is the (unique) UMVUE
- Can additionally *condition* on the selection or stopping rule used
- This gives the *uniformly minimum variance conditionally unbiased estimator* (UMVCUE)

Median unbiased estimator

- So far, have been considering *mean* unbiasedness (the usual unbiased property)
- A reasonable alternative is to consider *median* unbiasedness
- An estimator θ_{MU} is median unbiased if
$$Pr(\theta_{MU} \leq \theta) = Pr(\theta_{MU} \geq \theta)$$
 - ▶ i.e. the estimator underestimates just as often as it overestimates
- Median-unbiased estimators may be easier to construct than mean-unbiased estimators
- Median-unbiased estimators are invariant under one-to-one transformations (unlike mean-unbiased estimators)

Why not always use an unbiased estimator?

1. **Not (yet) available** for all classes of adaptive designs
2. **Increased MSE** (bias-variance trade-off) leading to large SEs and wide CIs

- Whitehead (1986) proposed a bias-adjusted MLE, which adjusts the MLE $\hat{\theta}_{\text{MLE}}$ to remove bias
- Bias for the MLE as a function of the unknown parameter of interest θ is $b(\theta) = E(\hat{\theta}_{\text{MLE}}|\theta) - \theta$
- Bias-adjusted MLE $\hat{\theta}_{\text{BC}}$ is the numerical solution to

$$\hat{\theta}_{\text{BC}} = \hat{\theta}_{\text{MLE}} - b(\hat{\theta}_{\text{BC}})$$

- **Group sequential designs:** Stevely et al. (2015) identified 68 trials that were published in leading medical journal
 - ▶ Only 7% (3/46) disclosed the use of some form of bias correction.

- **Group sequential designs:** Stevely et al. (2015) identified 68 trials that were published in leading medical journal
 - ▶ Only 7% (3/46) disclosed the use of some form of bias correction.
- Systematic search of adaptive designs in MEDLINE database, from 1st January 2000 to 30th May 2022

Type of adaptive design	Number of records screened	Number of randomised trials that reported results	Number of randomised trials that reported an unbiased or bias-adjusted estimate
MAMS	773	22	2
RAR	59	17	0
Adaptive enrichment	530	3	0

Why has there been so little uptake of adjusted estimators in practice?

- Belief that the bias of the MLE will typically be negligible in realistic trial scenarios
- Lack of awareness of the range of different unbiased and bias-adjusted estimators in the methodological literature
- Statistical software and code to calculate adjusted estimators is relatively sparse
- For more complex or novel adaptive designs, adjusted estimators may not exist

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Case study: MUSEC trial

Confidence intervals

Case study (continued)

Simulation study

Discussion

Case study: MUSEC trial

- Two-arm trial of oral cannabis extract vs placebo in adults with MS
- Primary endpoint: relief from muscle stiffness (yes/no) after 12 weeks
- Two-stage group-sequential design (maximum $n = 300$) with interim analysis (after $n = 200$)
 - ▶ O'Brien-Fleming efficacy stopping boundaries

	Interim data		Final data	
	Placebo	CE arm	Placebo	CE arm
Number of subjects with relief from muscle stiffness	12	27	21	42
Total number of subjects	97	101	134	143
Standardized test statistic	2.540		2.718	
OBF boundary	2.797		1.977	

Case study: MUSEC trial

Type of estimator	Estimator	Estimate (SE)	Relative difference
Standard/naive	MLE	0.137 (0.054)	–
Unconditional	MUE	0.134 (0.054)	–2%
	UMVUE	0.128 (0.054)	–7%
	Bias-corrected MLE	0.133 (0.055)	–3%
Conditional	MUE	0.185 (0.080)	+35%
	UMVCUE	0.172 (0.071)	+26%
	Bias-corrected MLE	0.191 (0.073)	+39%

MLE = Maximum Likelihood Estimator; MUE = Median Unbiased Estimator; UMVUE = Uniform Minimum Variance Unbiased Estimator; UMVCUE = Uniform Minimum Variance Conditionally Unbiased Estimator

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Coverage

- Suppose we have a random sample X from a probability distribution with single parameter θ of interest in the trial
- A CI for θ with confidence level $1 - \alpha$ is a random interval $(L(X), U(X))$ that has the following claimed property:
 $P(L(X) < \theta < U(X)) = 1 - \alpha$ for all θ
 - ▶ Note this is sometimes replaced by $P(L(X) < \theta < U(X)) \geq 1 - \alpha$ in the literature, particularly for discrete outcomes
- The coverage probability ('coverage') of a CI is given by $P(L(X) < \theta < U(X))$.
- The confidence level $1 - \alpha$ is the 'nominal' coverage probability
 - ▶ If all assumptions underlying CI are met, nominal coverage probability = (actual/true) coverage probability ('exact' coverage)
 - ▶ If assumptions not met, the actual coverage may be greater (overcoverage, the CI is a conservative CI) or less than (known as undercoverage, the CI is an anti-conservative CI) the nominal coverage probability

Potential Problems With Using Standard CIs for ADs

- 'Standard' CIs may not have the correct coverage (under or overcoverage).
 - ▶ Can lead to challenges around the interpretation of the study results, as well as use of CIs in secondary research such as evidence synthesis.
- Can be a disconnect between the test decision and the standard CI
 - ▶ Leads to substantial problems in interpretation and communication of the trial results

Desirable criteria for confidence intervals

1. Correct coverage probability [Essential!]
2. Narrower CIs are to be preferred
3. Consistent/compatible with the hypothesis test
4. Contains the point estimate of interest
5. Is an interval (i.e. not disjoint or the empty set)
6. Computational feasibility/simplicity

Classes of confidence intervals

- **Repeated CIs:** can be calculated at any stage, not tied to a pre-specified stopping rule, maintain simultaneous coverage. Can be conservative
- **Exact/monotone CIs:** can only be calculated at the stage a trial stops according to a pre-specified stopping rule, typically narrower on average than repeated CIs
- **Simultaneous CIs:** ‘multiplicity-adjusted’ individual CIs

Simultaneous Confidence Intervals

Multiplicity (multiple looks, multiple hypotheses) affects the construction of CIs just as it affects the significance levels of hypothesis tests

- Example: independent normal observations $X_1, \dots, X_{mK}$, each with mean θ and known variance σ^2
- Suppose after each group k of m observations, $k = 1, \dots, K$, we form the usual (unadjusted) 95% confidence interval $(\bar{X}_{mk} \pm 1.96\sigma/\sqrt{mk})$ for θ based on the observations accumulated so far
- The probability that all K intervals contain the true value of θ will be less than 95% (simultaneous coverage).

Simultaneous Confidence Intervals

Simultaneous coverage probabilities for unadjusted 95% CIs

Number of looks K	Overall probability that all K intervals contain θ
1	0.95
2	0.92
3	0.89
4	0.87
5	0.86
10	0.81
20	0.75
50	0.68
∞	0

(Jennison & Turnbull 2000)

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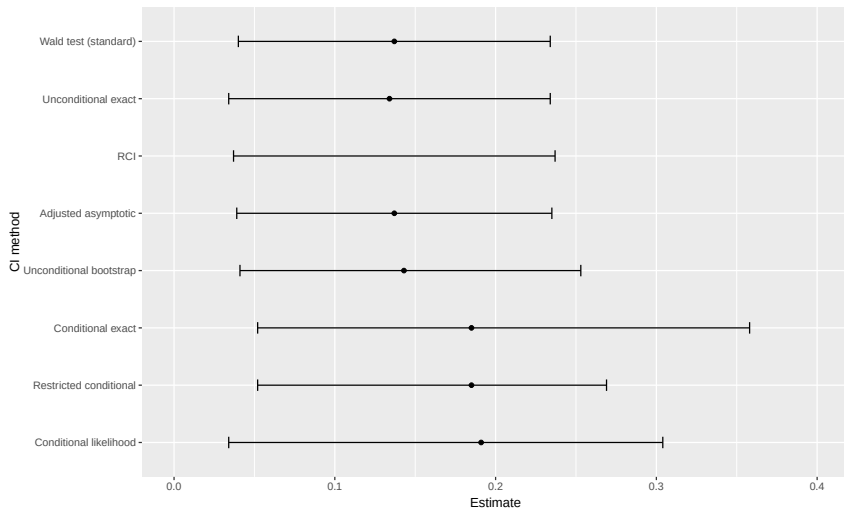
Discussion

Case study: MUSEC trial

Type of CI	CI method	Point estimate	95% CI	CI width
Standard/naive	Wald test	0.137 (MLE)	(0.040, 0.234)	0.194
Unconditional	Exact	0.134 (MUE)	(0.034, 0.234)	0.200
	Repeated	–	(0.037, 0.237)	0.199
	Adj. asymptotic	0.137	(0.039, 0.235)	0.196
	Bootstrap	0.143	(0.041, 0.253)	0.212
Conditional	Exact	0.185 (MUE)	(0.052, 0.358)	0.306
	Restricted Exact	0.185 (MUE)	(0.052, 0.269)	0.217
	Likelihood	0.191	(0.034, 0.304)	0.271

CI = Confidence Interval, MLE = Maximum Likelihood Estimator; MUE = Median Unbiased Estimator; RCI = Repeated Confidence Interval

Case study: MUSEC trial



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Simulation study: point estimation

- Run 10^5 simulations under different values of θ (0.10, 0.14, and 0.18), using the (asymptotic) joint distribution of the standardized test statistics at each stage

Type of estimator	Estimator	Difference in proportions (SE)		
		$\theta = 0.10$	$\theta = 0.14$	$\theta = 0.18$
MLE/naive	MLE (overall)	0.103 (0.054)	0.144 (0.054)	0.184 (0.053)
Unconditionally unbiased/bias-adjusted	MLE (stage 1)	0.100 (0.057)	0.140 (0.057)	0.180 (0.057)
	Median unbiased estimator (MUE)	0.101 (0.053)	0.142 (0.054)	0.182 (0.054)
	UMVUE	0.100 (0.052)	0.140 (0.054)	0.180 (0.055)
	Bias-corrected MLE (UBC-MLE)	0.101 (0.054)	0.142 (0.055)	0.183 (0.054)
Conditionally unbiased/bias-adjusted	MLE (stage 2)	0.100 (0.111)	0.140 (0.111)	0.180 (0.111)
	Conditional MUE (CMUE)	0.115 (0.083)	0.152 (0.080)	0.190 (0.081)
	UMVCUE	0.100 (0.062)	0.140 (0.071)	0.179 (0.080)
	Bias-corrected MLE (CBC-MLE)	0.111 (0.067)	0.154 (0.073)	0.194 (0.078)

Note: The probability of stopping at stage 1 was 0.15, 0.37 and 0.65 for $\theta = 0.10$, 0.14 and 0.18, respectively.

Simulation study: point estimation

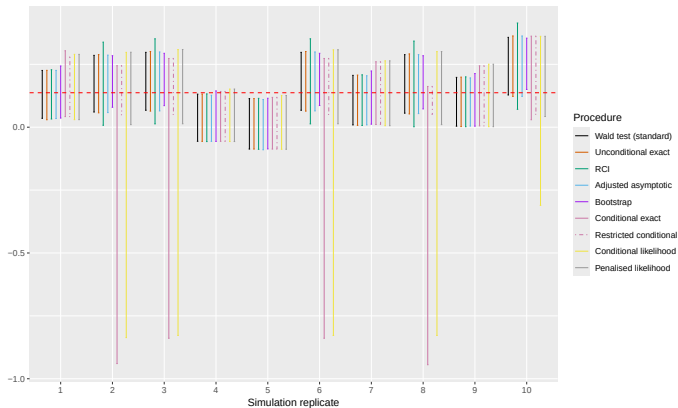
		Difference in proportions (SE)		
		$\theta = 0.10$	$\theta = 0.14$	$\theta = 0.18$
Trial stops early at the interim analysis	MLE (stage 1)	0.188 (0.025)	0.197 (0.031)	0.212 (0.038)
Trial continues to stage 2	MLE (overall)	0.087 (0.043)	0.113 (0.038)	0.132 (0.033)
	MLE (stage 1)	0.084 (0.045)	0.106 (0.038)	0.120 (0.030)
	Median unbiased estimator (MUE)	0.086 (0.041)	0.109 (0.034)	0.126 (0.027)
	UMVUE	0.084 (0.039)	0.106 (0.030)	0.120 (0.023)
	Bias-corrected MLE (UBC-MLE)	0.085 (0.041)	0.110 (0.036)	0.128 (0.032)

Note: The probability of stopping at stage 1 was 0.15, 0.37, and 0.65 for $\theta = 0.10$, 0.14, and 0.18, respectively.

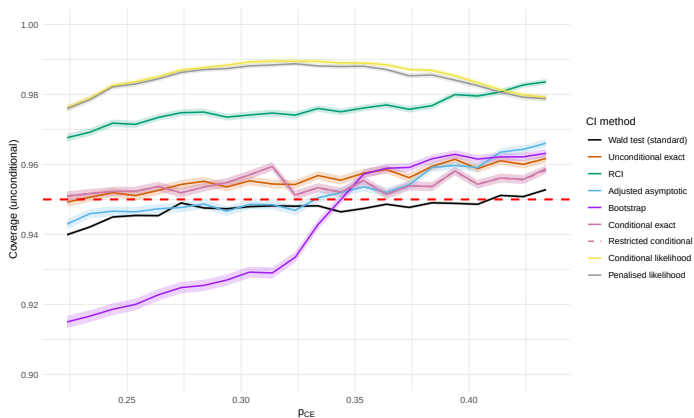
Simulation study: CIs

- Simulate repeated realisations of the MUSEC trial with true success rates (p_p, p_{CE}) on the two treatment arms
- Set $p_p = 21/134 \approx 0.157$ equal to the overall observed mean in the MUSEC trial
- Vary p_{CE} from 0.224 to 0.434
- Corresponds to probability of early stopping of 0.05 to 0.95
- 10^5 simulation replicates for each pair of parameter values

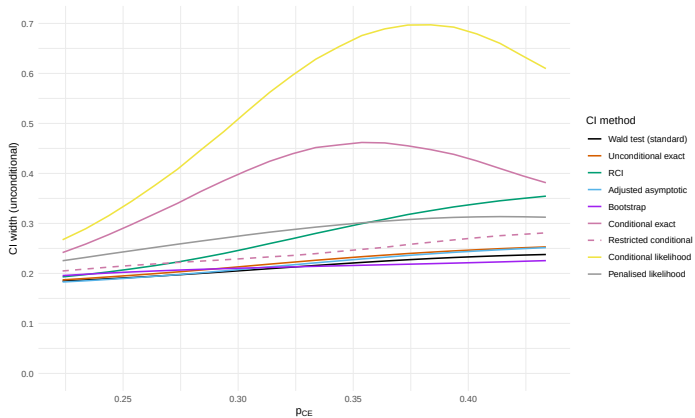
Realised CIs



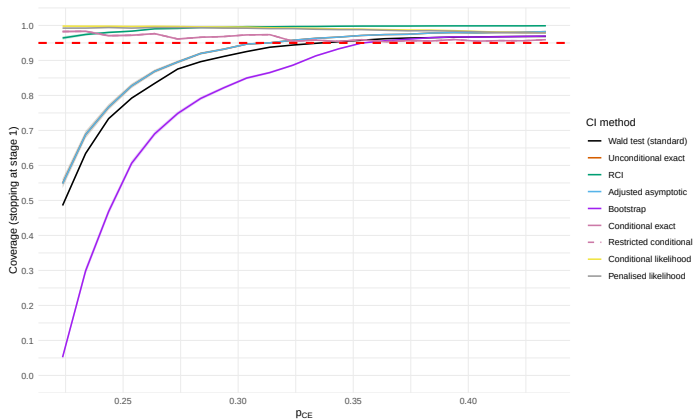
Coverage (unconditional)



CI width (unconditional)



Coverage (conditional on stopping at stage 1)



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How to do estimation in practice

- The issue of estimation should be considered throughout the whole lifecycle of an adaptive trial
- The design and analysis of an adaptive trial are closely linked, and one should not be considered without the other
- Context, aims and design of an adaptive trial should all inform the analysis strategy used, which includes the choice of point estimators and CIs
- A review of the literature may be sufficient to compare their performance
- Otherwise, conduct simulations to explore their properties
- Statistical analysis plan should include a description and justification of the point estimators and CIs used

Planning stage

- Decide on what exactly is to be estimated (i.e. the estimands of interest)
- Decide on desired characteristics of point estimators and CIs:
 - ▶ Conditional versus unconditional perspective (see Marschner 2021 and tomorrow's lecture)
 - ▶ Bias-variance trade-off for point estimators
 - ▶ Trade-off in metrics for CIs
 - ▶ Link between CIs and point estimators
- A review of the literature may be sufficient. Otherwise, conduct simulations to explore

Pre-specification of analyses

- Statistical analysis plan (SAP) and health economic analysis plan (HEAP) should include a description and justification of the estimators and CIs used
- When available, unbiased or bias-reduced estimators should be used and reported alongside the standard MLE
- If multiple adjusted estimators/CIs are available and are of interest, one adjusted estimator/CI should be designated the 'primary' adjusted estimator/CI
 - ▶ Others included as sensitivity or supplementary analyses

Reporting results for a completed trial

- There should be a clear description of the statistical methods used to estimate treatment effects
- Adjusted point estimates and CIs taking the trial design into account are to be preferred (see FDA, EMA and ICH E20 guidance!)

Review papers: point estimation for ADs

Received: 25 June 2021 | Revised: 21 October 2022 | Accepted: 1 November 2022
DOI: 10.1002/sim.9605

RESEARCH ARTICLE

Statistics
in Medicine WILEY

Point estimation for adaptive trial designs I: A methodological review

David S. Robertson¹ | Babak Choodari-Oskooei² | Munya Dimairo³ |
Laura Flight³ | Philip Pallmann⁴ | Thomas Jaki^{1,5}

Received: 25 June 2021 | Revised: 20 January 2023 | Accepted: 18 March 2023
DOI: 10.1002/sim.9734

RESEARCH ARTICLE

Statistics
in Medicine WILEY

Point estimation for adaptive trial designs II: Practical considerations and guidance

David S. Robertson¹ | Babak Choodari-Oskooei² | Munya Dimairo³ | Laura Flight³ |
Philip Pallmann⁴ | Thomas Jaki^{1,5}

- Two-part paper series in *Statistics in Medicine*
 - ▶ **Part I:** Review methods for unbiased and bias-reduced estimation of treatment effects after an adaptive clinical trial and critically discuss different approaches
 - ▶ **Part II:** Point estimation for adaptive designs from a practical perspective, including a set of guidelines for best practice
- Available at doi:10.1002/sim.9605 and doi:10.1002/sim.9734

Review papers: confidence intervals for ADs

Statistics in Medicine

WILEY
Statistics
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REVIEW ARTICLE **OPEN ACCESS**

Confidence Intervals for Adaptive Trial Designs I: A Methodological Review

David S. Robertson¹ | Thomas Burnett² | Babak Choodari-Oskooei³ | Munya Dimairo⁴ | Michael Grayling⁵ | Philip Pallmann⁶ | Thomas Jaki^{1,7}

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REVIEW ARTICLE **OPEN ACCESS**

Confidence Intervals for Adaptive Trial Designs II: Case Study and Practical Guidance

David S. Robertson¹ | Thomas Burnett² | Babak Choodari-Oskooei³ | Munya Dimairo⁴ | Michael Grayling⁵ | Philip Pallmann⁶ | Thomas Jaki^{1,7}

¹MRC Biostatistics Unit, University of Cambridge, Cambridge, UK | ²University of Bath, Bath, UK | ³MRC Clinical Trials Unit at UCL, London, UK | ⁴Sheffield Centre for Health and Related Research (SCHARR), University of Sheffield, Sheffield, UK | ⁵Statistics and Decision Sciences, James Ridd, High Wycombe, UK | ⁶Centre for Trials Research, Cardiff University, Cardiff, UK | ⁷University of Regensburg, Regensburg, Germany

Correspondence: David S. Robertson (david.robertson@mrc-bu.cam.ac.uk)

- Two-part paper series in *Statistics in Medicine*
 - ▶ **Part I:** Review methods for adjusted confidence intervals for adaptive trials and critically discuss different approaches
 - ▶ **Part II:** Illustrates the computation of different CIs in practice, including a set of guidelines for best practice
- Available at doi:10.1002/sim.70174 and doi:10.1002/sim.70202

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Robertson DS et al. (2025b)

Confidence intervals for adaptive trial designs II: Practical considerations and guidance
Statistics in Medicine, 44 (18-19), e70202

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An Investigation of the Shortcomings of the CONSORT 2010 Statement for the Reporting of Group Sequential Randomised Controlled Trials: A Methodological Systematic Review.
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